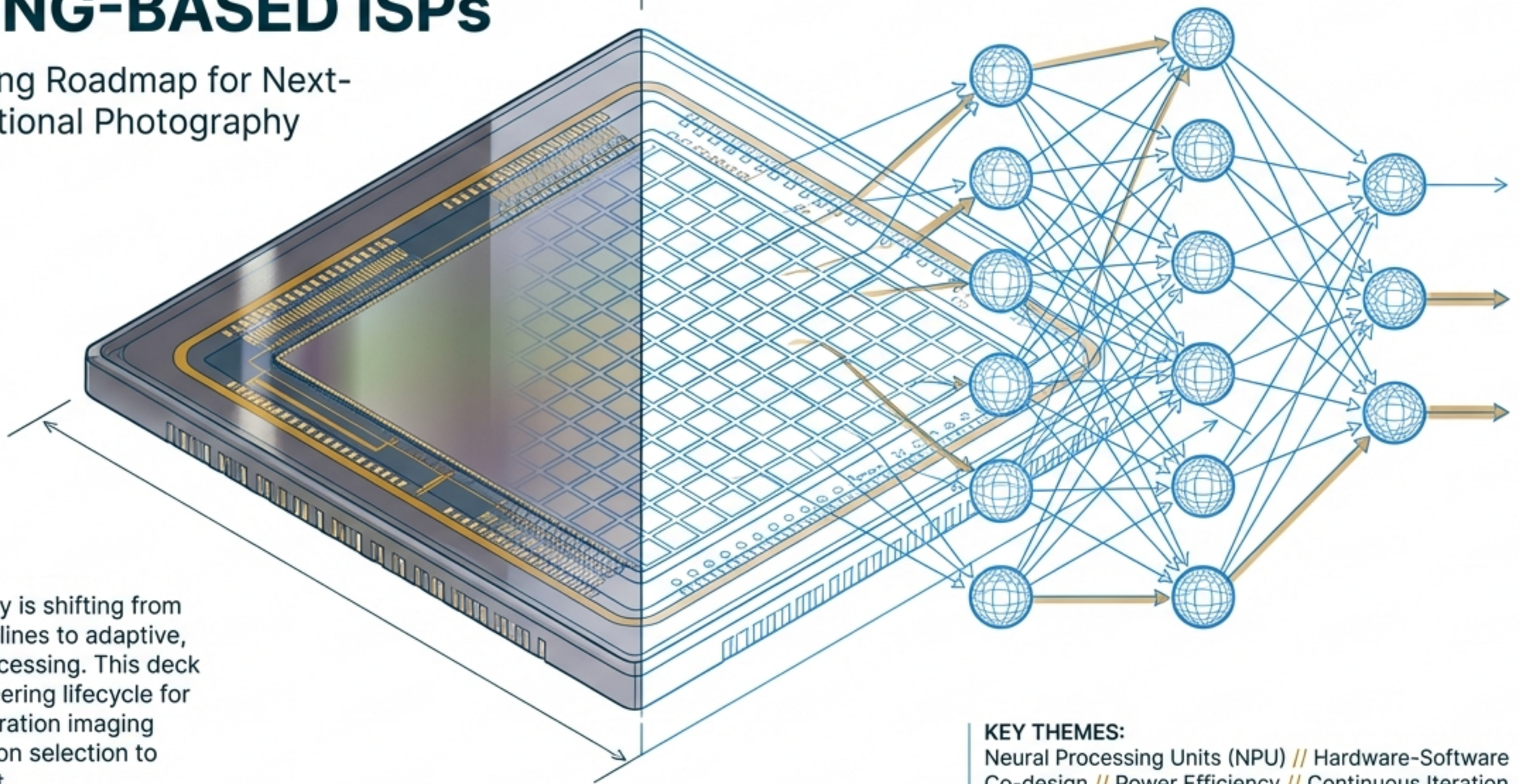
ARCHITECTING DEEP LEARNING-BASED ISPs

The Engineering Roadmap for Next-Gen Computational Photography



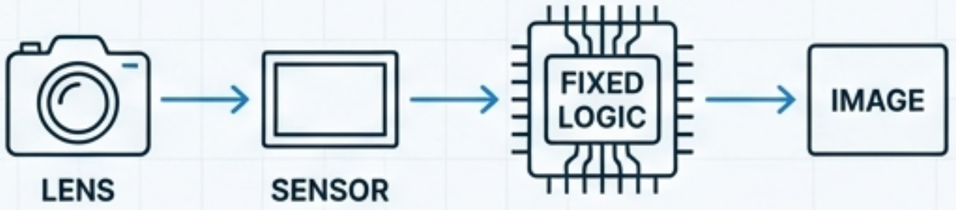
Mobile photography is shifting from fixed-function pipelines to adaptive, AI-accelerated processing. This deck outlines the engineering lifecycle for building next-generation imaging systems, from silicon selection to driver development.

KEY THEMES:
Neural Processing Units (NPU) // Hardware-Software Co-design // Power Efficiency // Continuous Iteration

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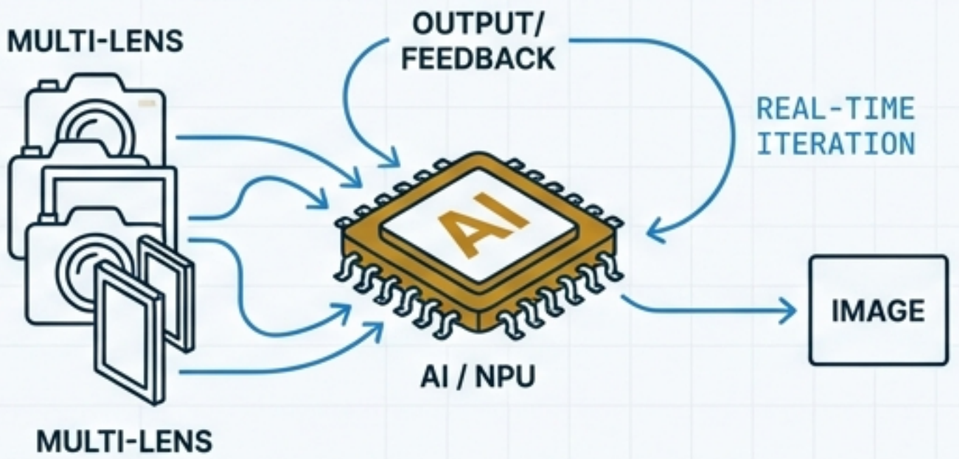
THE PARADIGM SHIFT IN IMAGE SIGNAL PROCESSING

TRADITIONAL ISP



Defined by fixed pipelines. Limited flexibility.
Struggles with complex scene understanding.

DEEP LEARNING ISP



Real-time high-res processing. AI accelerators.
Multi-camera fusion.



Scene Recognition
Intelligent environmental adjustment.



Super-Resolution
AI-driven detail enhancement.



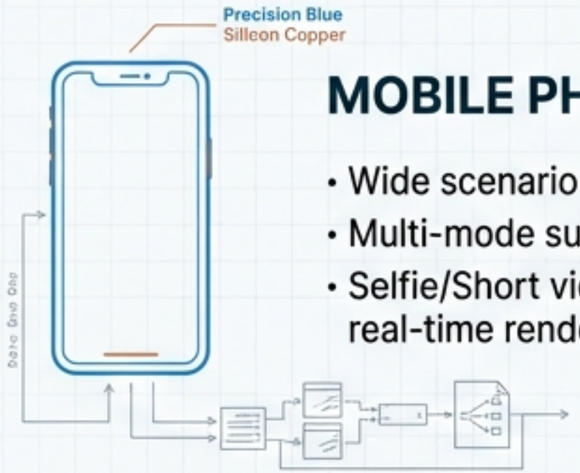
Noise Reduction
Deep learning for low-light.



Object Detection
Real-time focus tracking.

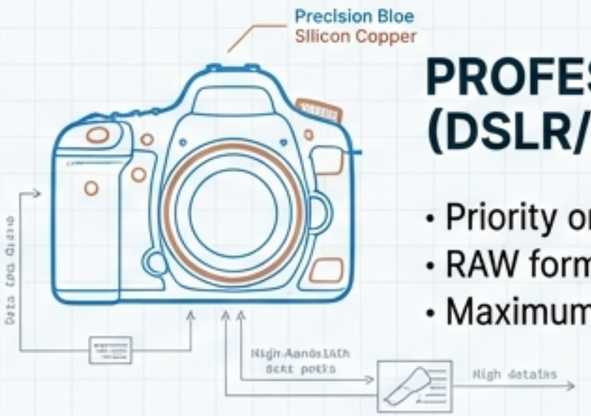
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PHASE I: ANALYZING MARKET REQUIREMENTS AND SCENARIOS



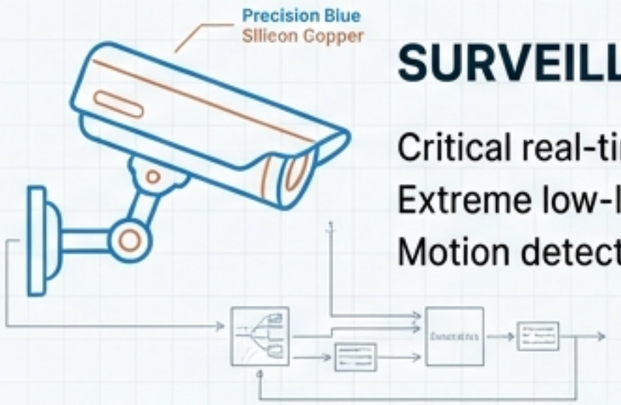
MOBILE PHONES

- Wide scenario coverage (low-light to bright).
- Multi-mode support (Night, Portrait).
- Selfie/Short video optimization requiring real-time rendering.



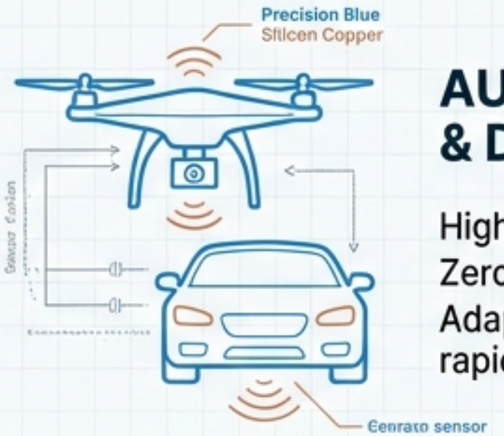
PROFESSIONAL CAMERAS (DSLR/MIRRORLESS)

- Priority on high resolution.
- RAW format support.
- Maximum dynamic range and color fidelity.



SURVEILLANCE SYSTEMS

Critical real-time stream processing.
Extreme low-light performance.
Motion detection and facial recognition.



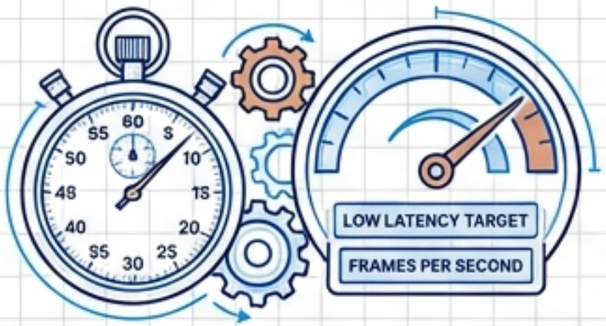
AUTONOMOUS DRIVING & DRONES

High reliability.
Zero-latency requirements.
Adaptability to varying altitudes and rapid lighting changes.

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Defining the Performance Envelope

PROCESSING SPEED



Low latency target. Frames per second.
Critical for real-time video.

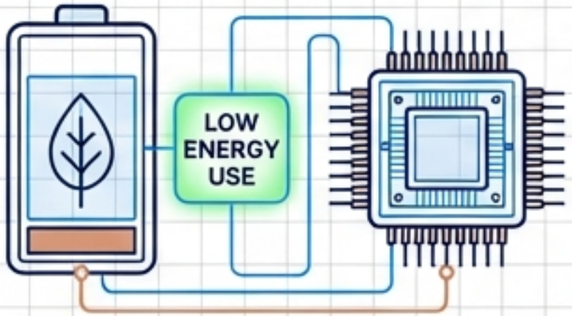
IMAGE QUALITY



Resolution, Clarity, Color Accuracy,
Low Noise (SNR).

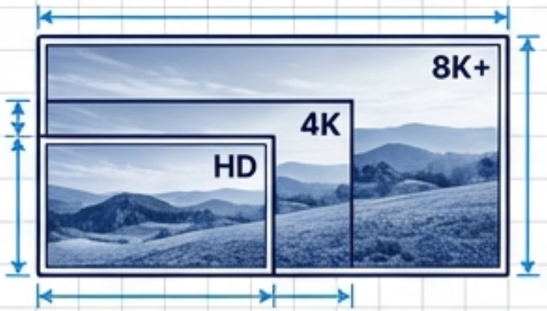
ISP
PERFORMANCE METRICS DEFINITION

POWER CONSUMPTION



Low Energy Use. Mobile Efficiency
& Battery Life. Thermal management.

SUPPORTED RESOLUTION

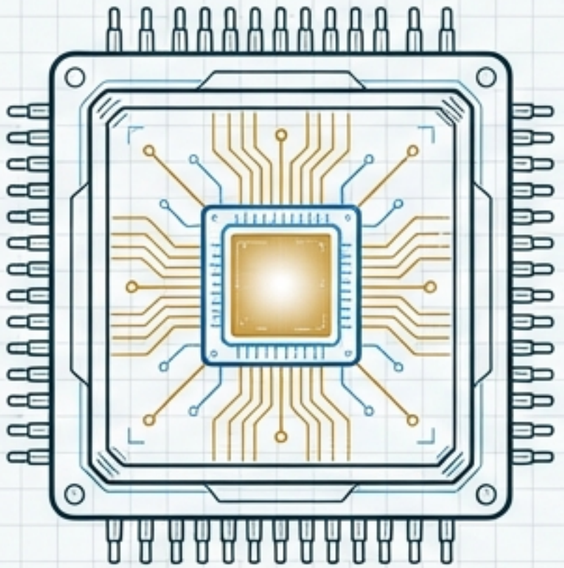


High-Res Image Handling.
8K+ support without speed compromise.

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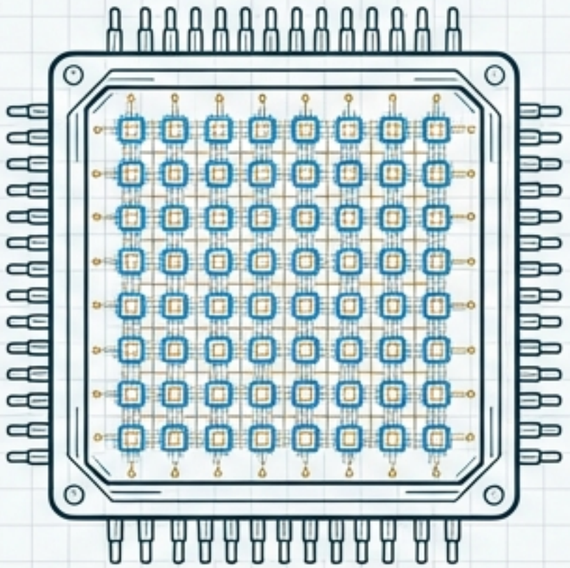
Phase II: Phase II: Selecting the Processing Unit

Central Processing Unit (CPU)



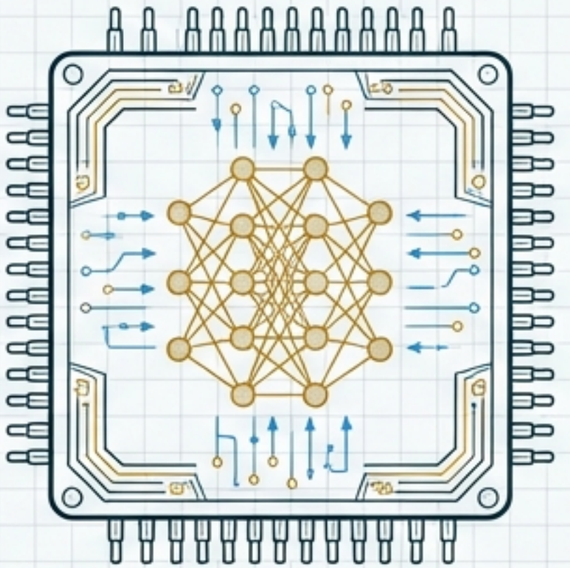
Role: General purpose, complex logic, control processes.
Limitation: Less efficient for high concurrency and massive parallel data.

Graphics Processing Unit (GPU)



Role: High concurrency, parallel computing.
Strength: Accelerating deep learning models and high-res image/video.

Dedicated AI Accelerator (NPU/TPU)

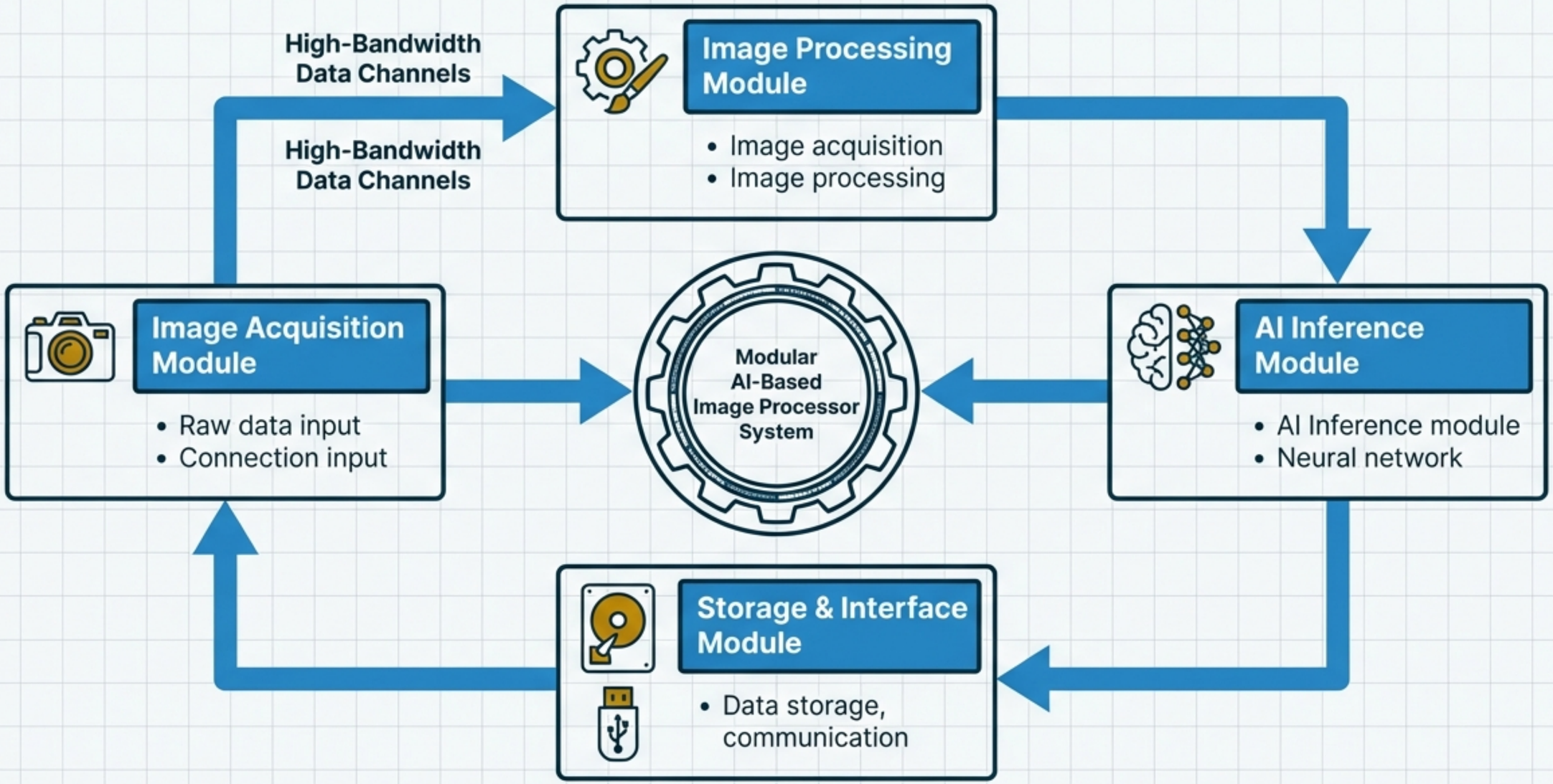


Role: Specialized hardware for matrix operations/tensor processing.
Advantage: Most efficient for inference; boosts recognition and classification.

Decision Goal → **Achieve the best balance of Performance vs. Efficiency.**

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System Architecture and Data Flow



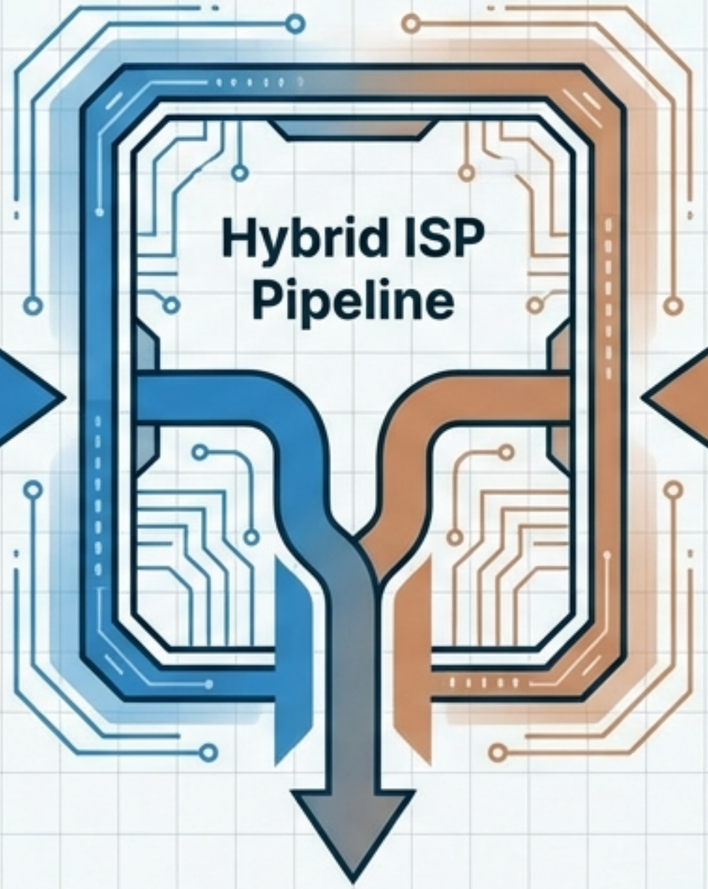
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Algorithm Development Strategy: The Hybrid Model

Traditional Algorithms

- Demosaicing
- Color Correction
- White Balance

Strength: Stability & Reliability for foundational image construction.



AI Algorithms

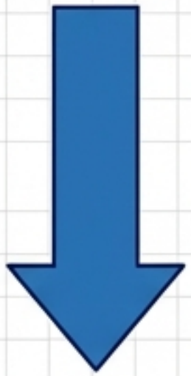
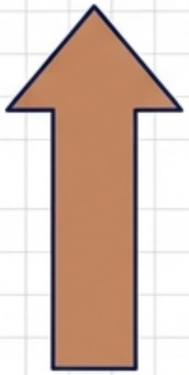
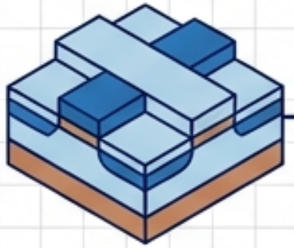
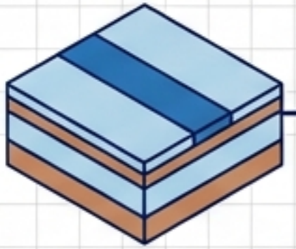
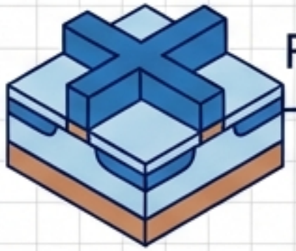
- Image Recognition
- Semantic Enhancement
- Super-Resolution

Strength: Handling complex, non-linear problems.

Optimization: Techniques like Model Pruning and Quantization are required to fit AI models onto mobile silicon.

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Phase III: Semiconductor Manufacturing Choices

Process Nodes (Technology)	**Process Type** (Architecture)	**Foundry** (Manufacturing Partner)
 14nm 7nm 5nm  Transistor Density & Performance  Cost	 CMOS  Smartphone/IoT  SOI  Rocket/Satellite  FinFET  Server/GPU	tsmc SAMSUNG intel Global Fabrication Capacity & Expertise

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Integrated Circuit Design Workflow

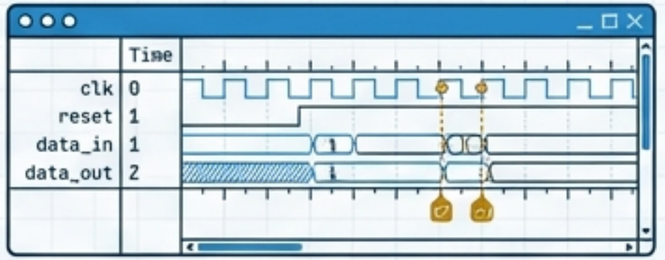
Hardware Description Language (HDL)

```
// Code Verilog Process
// VHDL syntax
module counter_dod(
  input clk, reset,
  output [2:8] count
);
  // .. Verilog ...
endmodule

// Code VHDL
entity counter_vhdl is
  // ... VHDL ...
end entity;
```

Verilog VHDL

Logic Verification

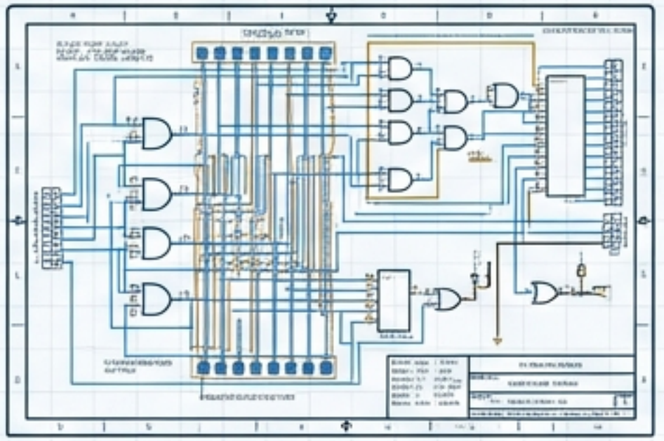


Functional Sim Timing Sim Formal Verif

Design Tools (EDA)

cadence SYNOPSYS Mentor Graphics

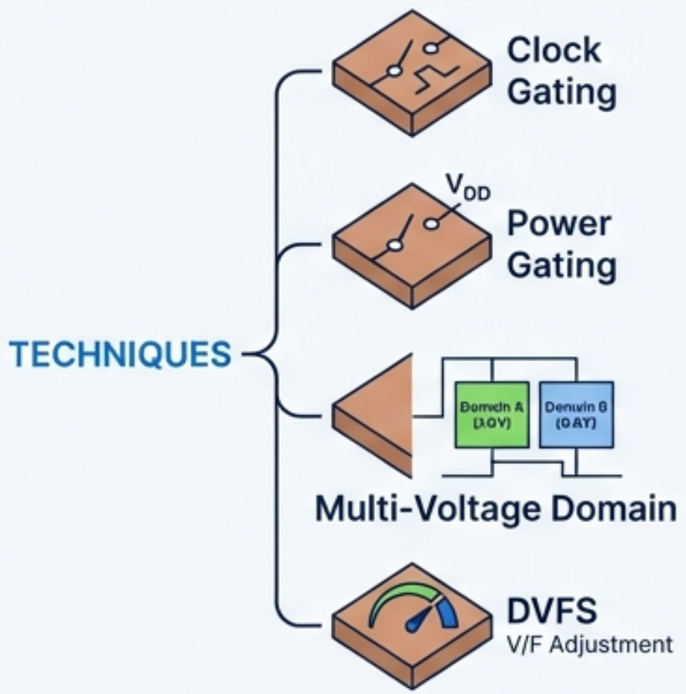
Place & Route



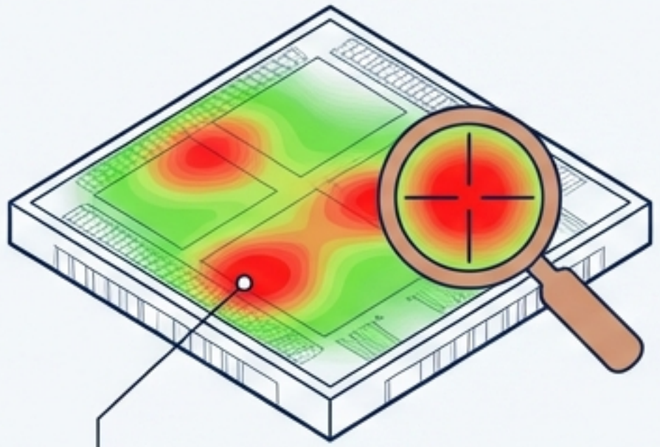
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Power Management and Optimization

1. Low-Power Design Techniques

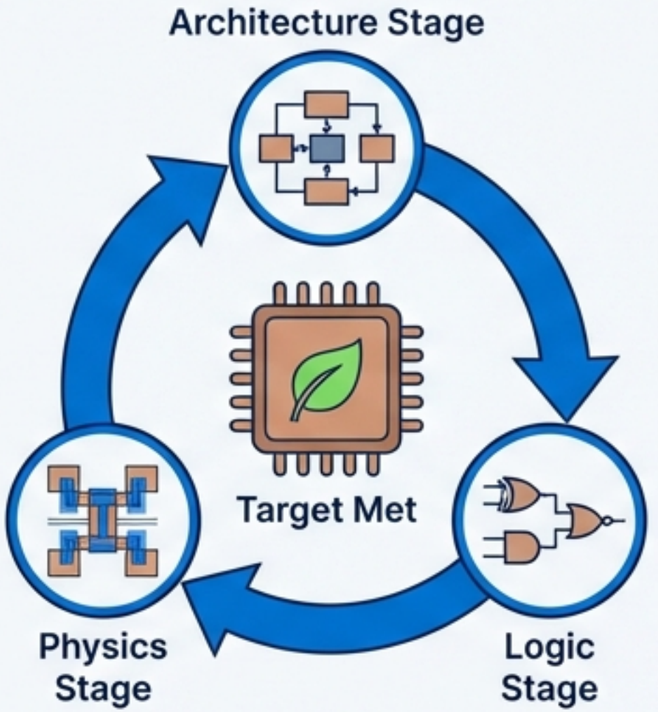


2. Power Analysis



Identify bottlenecks & high-consumption areas.

3. Power Optimization (Iterative Improvement)



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Testing and Validation Methodology



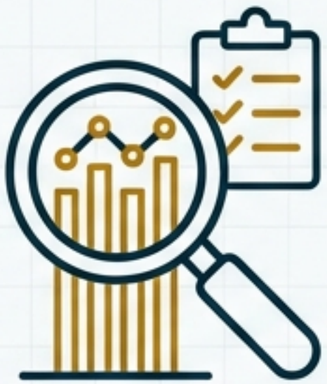
Test Environment Setup

Mimic real-world use cases (Low-light, motion).



Data Collection

Real-time image/video capture. Record power/timing.



Data Analysis Methods

Statistical analysis vs. Subjective scoring.



Problem Identification

Identify deficiencies. Analyze degraded quality.

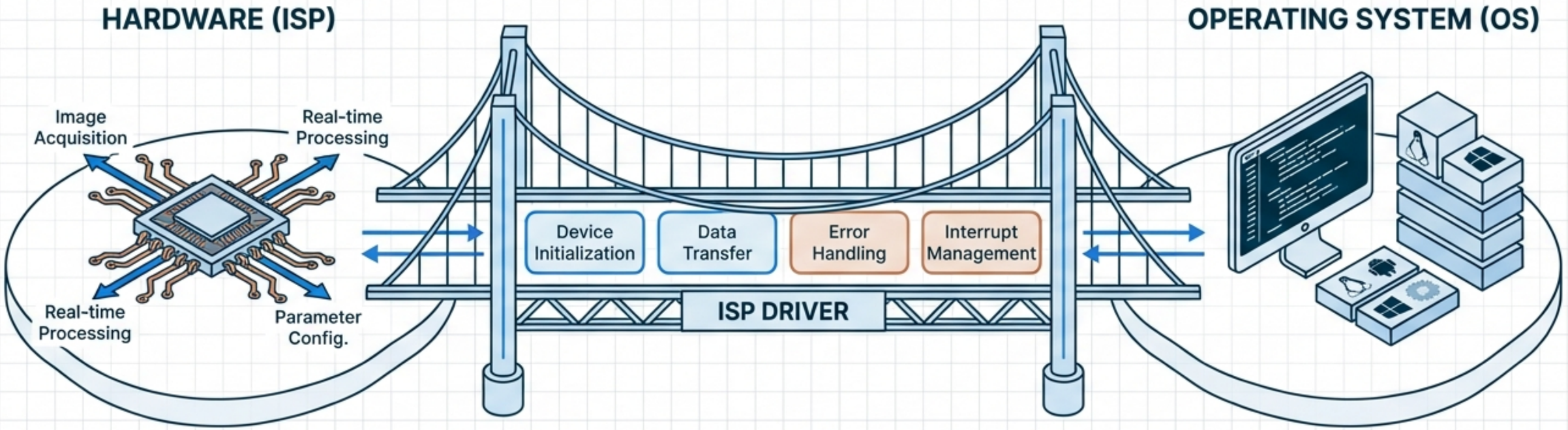


Summary & Report

Compile results. Support design optimization.

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Phase IV: The Software Bridge (Driver Development)

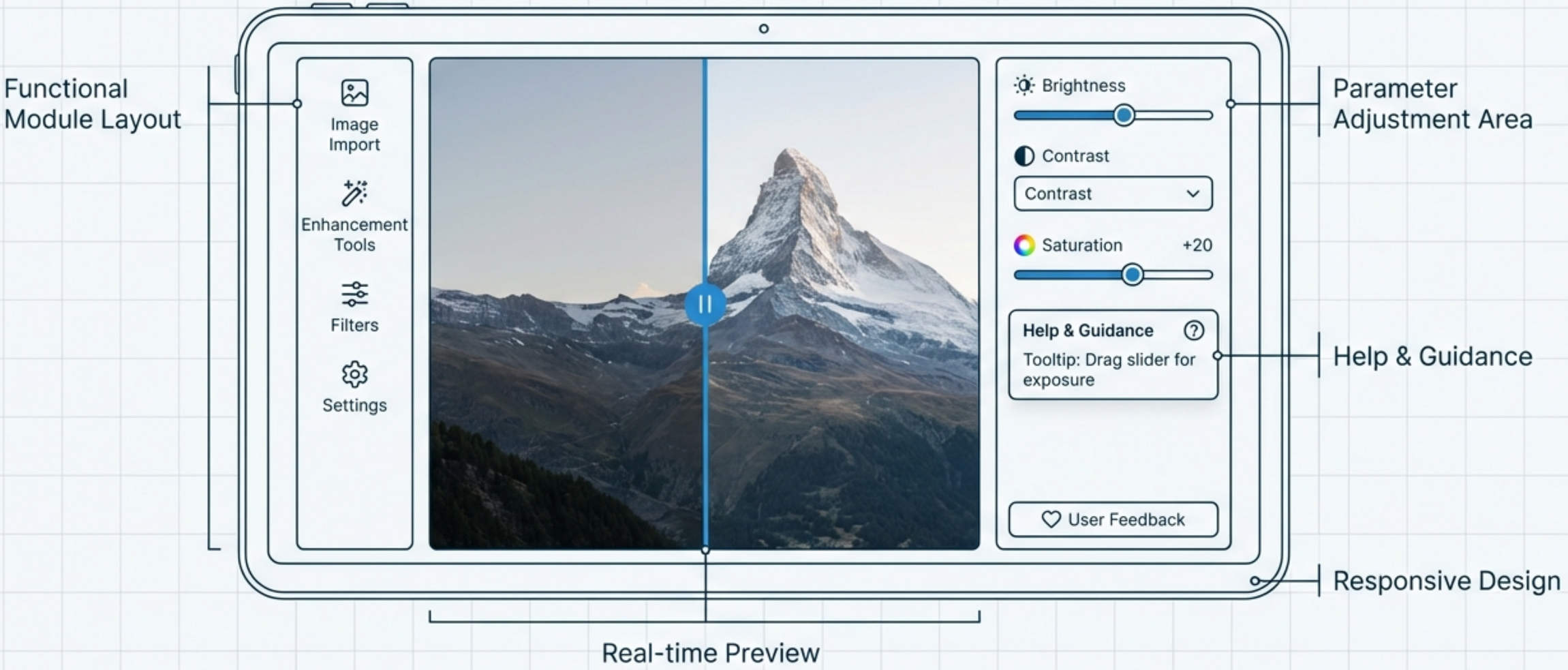


Driver Development Process

1. Architecture Design
2. Hardware Interface Definition
3. Function Implementation
4. System Integration
5. Documentation

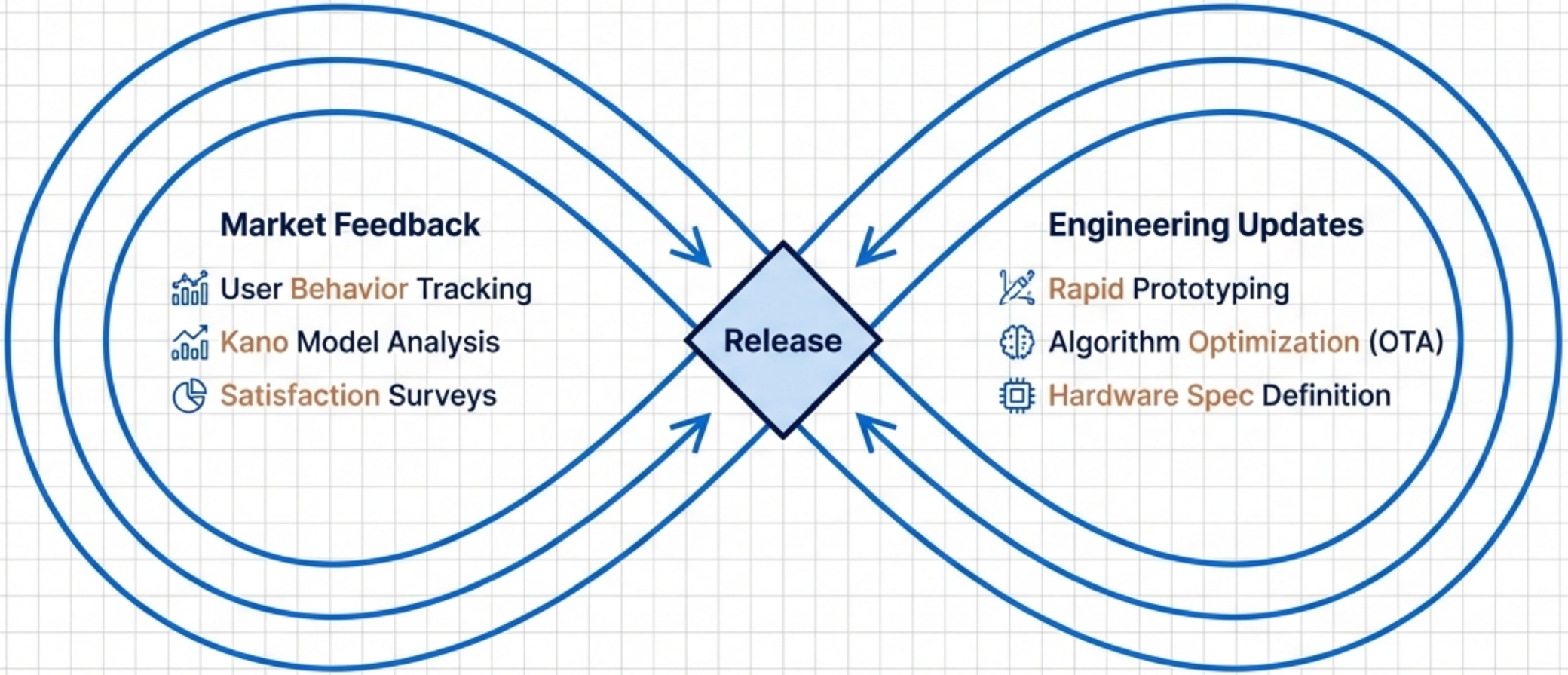
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User Interface and Experience Design



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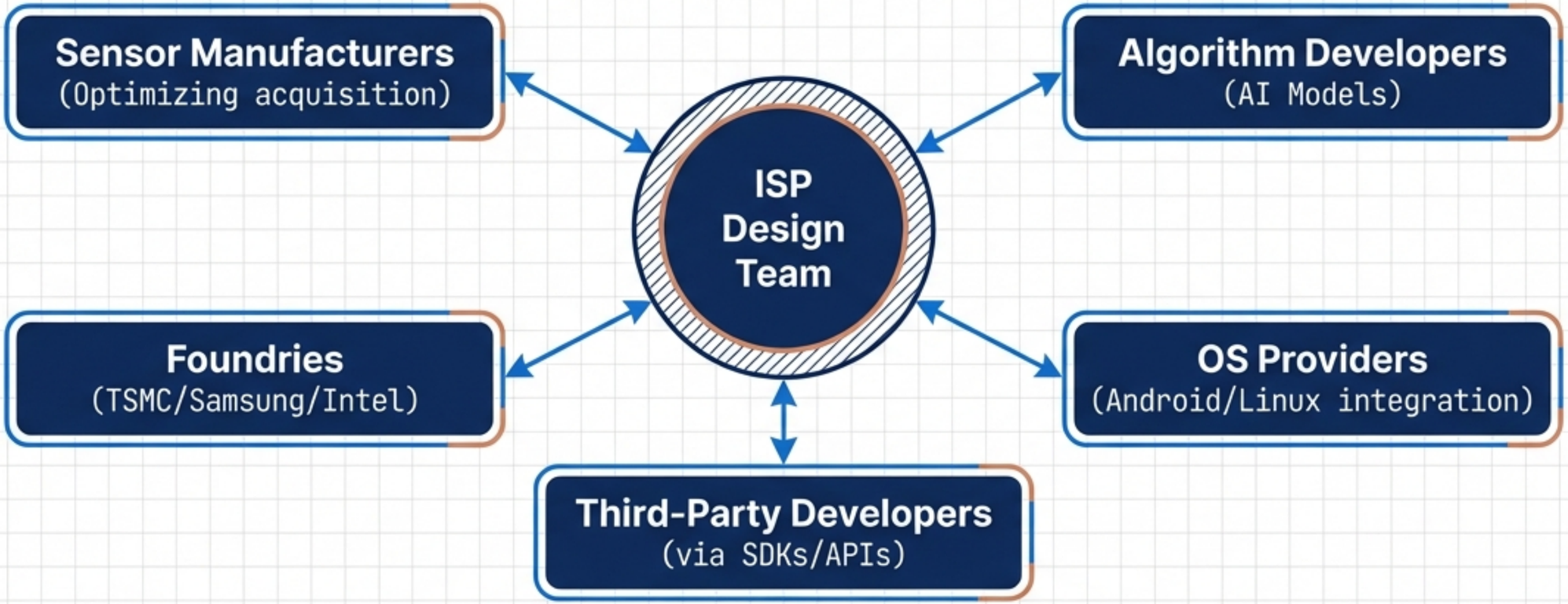
Iteration and Continuous Improvement



The Goal: Establish a circular feedback mechanism where market data directly informs the next generation of silicon and software.

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Collaboration & Ecosystem



Future Trends

1. Zero-Shutter-Lag AI Processing • 2. Tighter NPU/ISP Integration • 3. 3nm Process Nodes

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